

A Multi-Level Intelligent Detection Model for Malicious URLs and HTML-Based Web Attacks Using Machine Learning

Abstract: Malicious URLs and compromised web pages are major sources of cyberattacks such as phishing, malware distribution, and data theft. Existing systems rely mainly on blacklist-based filtering and static rules, which are ineffective against zero-day attacks. This project proposes a multi-level intelligent detection model that analyzes URL features using machine learning and inspects HTML content for hidden malicious scripts. The results are combined to automatically block and revoke access to malicious URLs, improving detection accuracy and web security.